Xiaotian Hu

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Bio

A strongly self-motivated third-year undergraduate student majoring in Biomedical Engineering at Beihang University, currently ranked first in the major. Admitted under Beihang's broad Aerospace Engineering category, I ranked 1/416 in my first year and subsequently chose BME as my major. I am a two-time recipient of the National Scholarship and have earned multiple distinctions in academic and research competitions. My research interests include AI agents, multimodal vision-language models, and large language models.

Academic Background

- **Beihang University (BUAA)** *Sep 2023 – Expected Jun 2027*
  Bachelor of Engineering in Biomedical Engineering Beijing, China
  - **Academic Performance:**
    - **Biomedical Engineering** (Sep 2023 – Feb 2026): Average Score: 94.04/100; Rank: 1/50
    - **General Engineering (Aerospace Track)** (Sep 2023 – Jun 2024): Average Score: 93.54/100; Rank: 1/416
  - **Awards:** Two-time National Scholarship (top 0.2%), Merit Student Award, and Outstanding Student Award.
  - **Selected Coursework:**
    - Linear Algebra (100), Numerical Analysis (100), Engineering Graphics (100), Fundamental Physics B (100)
    - Mathematical Analysis for Engineering (98), Probability and Statistics (97), Methods of Mathematical Physics (96)
  - **Technical Skills:** Python, C, PyTorch, MATLAB, SolidWorks, and LaTeX.
  - **English:** CET-6: 537; CET-4: 529.

Publications

Published / Accepted

1. **X. Hu**, J. Huang, M. Liu, et al. FetalAgents: A multi-agent system for fetal ultrasound image and video analysis. ***International Conference on Medical Image Computing and Computer Assisted Intervention***. **(MICCAI 2026 Early Accept, Top 9%)**.[Paper][Code]
2. **X. Hu**, et al. Benchmarking spatiotemporal fetal brain atlas construction. ***Organization for Human Brain Mapping Annual Meeting***. (OHBM 2026 Poster).
3. **X. Hu**, et al. INSTA: Implicit neural spatio-temporal atlas from thick-slice clinical fetal brain MRI. ***ISMRM Workshop on Unlocking the Potential of Prenatal MRI: Advances in Fetal Brain, Heart & Placenta Imaging***. (ISMRM 2026 Oral).
4. Z. Gao, et al., **X. Hu**, and Q. Tian. View-Conditioned Semi-Supervised Learning for Fetal Cardiac Ultrasound Analysis. ***IEEE 23rd International Symposium on Biomedical Imaging***. (ISBI 2026).[Paper]
5. Y. Li, M. Liu, L. Chang, et al., **X. Hu**, et al. Maximizing Domain Generalization in Automated Fetal Brain Biometry. ***International Conference on Medical Image Computing and Computer Assisted Intervention***. (MICCAI 2026).[Code]

Under Review / Submitted

1. **X. Hu**, M. Liu, J. Huang, et al. Towards reliable fetal ultrasound interpretation with multi-agent collaboration. ***Medical Image Analysis***. (MedIA, Major Revision).[Paper][Code]
2. **X. Hu**, M. Liu, H. Yang, et al. INFANiTE: Implicit neural representation for high-resolution fetal brain spatio-temporal atlas learning from clinical thick-slice MRI. ***Proceedings of the AAAI Conference on Artificial Intelligence***. (AAAI 2027, Under Review).[Paper][Code]
3. H. Yang, M. Liu, Y. Hao, **X. Hu**, et al. Simple Baselines for Fetal Brain Unsupervised Anomaly Detection. ***IEEE International Conference on Bioinformatics and Biomedicine***. (BIBM 2026, Under Review).[Code]
4. Y. Luo, Y. Li, M. Liu, et al., **X. Hu**, et al. FetAngle: Toward Generalizable Automated Fetal Brain Angle Biometry via Test-Time Adaptation. ***British Machine Vision Conference***. (BMVC 2026, Under Review).[Code]
5. V. Galinova, M. Seifi, et al., **X. Hu**, et al. AI4Life Open Calls and Public Challenges: Why, How, and What We Have Learned. ***npj Imaging***. (*Submitted*).

Research Experience

- **Research Assistant – BIRTH Lab, Tsinghua Biomedical Engineering** *Advisor: Qiyuan Tian*
  - **Fetal VQA**: Built the first dedicated fetal-ultrasound visual question answering benchmark, together with a multi-dataset evaluation suite spanning seven clinical tasks.
  - Developed *FetalAgents*, a tool-augmented multi-agent framework for fetal ultrasound visual question answering, report generation, image captioning, and video summarization.
  - **MICCAI Early Accept:** *FetalAgents: A Multi-Agent System for Fetal Ultrasound Image and Video Analysis*.
  - **MIA Major Revision:** *Towards Reliable Fetal Ultrasound Interpretation with Multi-Agent Collaboration*.
  - **INFANiTE**: Proposed an implicit-neural-representation framework for constructing high-resolution spatio-temporal fetal brain atlases directly from clinical thick-slice MRI, reducing the pipeline from days to hours.
  - **AAAI Under Review:** *INFANiTE: Implicit Neural Representation for High-Resolution Fetal Brain Spatio-Temporal Atlas Learning from Clinical Thick-Slice MRI*.
- **Algorithm Engineer – FacePhys & HCI Lab, Tsinghua Computer Science** *Mentor: Jiankai Tang*
  - **Human Foundation Model:** Contributed to a human-centric multimodal streaming foundation model supporting real-time full-duplex interaction.
  - Built data collection pipelines to collect facial videos and livestreams from YouTube and Twitch for model training.
  - Investigated algorithms for blood-pressure estimation from facial videos.
- **Visiting Student – Electrical and Computer Engineering, HKU** *Advisor: Cheng Chen*
  - Validated a medical vision-language model with Euclidean, hyperbolic, and spherical representations, demonstrating strong downstream performance.
  - **NMI Manuscript in Preparation:** Second-author work targeting *Nature Machine Intelligence*.

Competitions and Challenges

- **Biomedical Engineering Conference and Innovation Medical Summit** *Poster Presenter*
  China Apr 2026
  - Poster accepted for presentation at the national conference, organized by the Chinese Society of Biomedical Engineering.
- **AI4Life Supervised Microscopy Image Denoising Challenge** *Participant & Co-author*
  International Sep 2025
  - Achieved **Second Place** in the AI4Life Supervised Microscopy Image Denoising Challenge.
  - **Co-author:** *AI4Life Open Calls and Public Challenges: Why, How, and What We Have Learned*, submitted to *npj Imaging*.
- **16th Chinese Mathematics Competition for College Students** *Participant*
  China Dec 2024
  - Awarded **First Prize**.
- **40th National College Physics Competition (Selected Regions)** *Participant*
  China Dec 2024
  - Awarded **Third Prize**.
- **Mathematical Contest in Modeling** *Participant*
  International May 2024
  - Awarded **Meritorious Winner**.